

Accenture Network & Security Exam Preparation Notes

Comprehensive Technical Assessment Question Bank with Complete Option-by-Option Breakdown

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Category: IT All Companies / Networking
Topics

Last Modified: February 15, 2026

Assessment Overview: This guide provides a complete, structured compilation of frequently tested questions from the Accenture Technical Assessment. Core modules covered include *Basics of Networking*, *Network Security*, *Encryption Standards & Algorithms*, *Network Security Devices*, *Firewalls*, and *Attack Types*. Every question includes in-depth analyses of all choices to ensure thorough conceptual mastery.

1. BASICS OF NETWORKING

Q1 Which protocol is responsible for assigning IP addresses dynamically in a network?

- | | |
|---------|---------|
| A. HTTP | B. FTP |
| C. DHCP | D. ICMP |

Correct Answer: C. DHCP

Detailed Option Breakdown:

- A. HTTP (HyperText Transfer Protocol):** Application Layer protocol (Port 80) designed for distributing hypermedia and web pages. It does not handle IP addressing.
- B. FTP (File Transfer Protocol):** Client-server protocol (Ports 20/21) for transferring files between network nodes; does not allocate network configuration.
- C. DHCP (Dynamic Host Configuration Protocol):** **[CORRECT]** Automatically assigns IP addresses, subnet masks, default gateways, and DNS servers dynamically using the 4-step *DORA* process (Discover, Offer, Request, Acknowledge) over UDP ports 67/68.
- D. ICMP (Internet Control Message Protocol):** Layer 3 protocol used by network devices for diagnostic reporting and error signaling (e.g., ping, traceroute).

Q2 Which layer of the OSI model is responsible for reliable communication and flow control?

- | | |
|--------------|----------------|
| A. Transport | B. Network |
| C. Data Link | D. Application |

Correct Answer: A. Transport

Detailed Option Breakdown:

- A. Transport (Layer 4):** **[CORRECT]** Coordinates end-to-end data transfer, manages segmentation, ensures connection reliability, prevents buffer overload via flow control (sliding window), and performs error recovery (TCP).
- B. Network (Layer 3):** Handles logical IP addressing, packet forwarding, and path selection across intermediate routers.
- C. Data Link (Layer 2):** Deals with node-to-node framing, physical MAC addresses, and local media access control (CSMA/CD).
- D. Application (Layer 7):** Directly interfaces with end-user software applications (browsers, email clients).

Q3 What is the main function of a router in a network?

- | | |
|--|---|
| A. Broadcast packets | B. Block unauthorized access |
| C. Forward data packets between networks | D. Translate domain names to IP addresses |

Correct Answer: C. Forward data packets between networks

Detailed Option Breakdown:

- **A. Broadcast packets:** Routers break broadcast domains by default; broadcasting to all ports is the role of Layer 1 Hubs or unmanaged switches.
- **B. Block unauthorized access:** While routers can enforce basic ACLs, traffic inspection and blocking is primarily the function of dedicated Firewalls.
- **C. Forward data packets between networks:** **[CORRECT] Routers are Layer 3 devices that interconnect different logical subnets and direct packets along the optimal path using routing tables (OSPF, BGP, RIP).**
- **D. Translate domain names to IP addresses:** This is performed exclusively by DNS (Domain Name System) servers.

Q4 Which of the following is a valid IP address in IPv4 format?

- | | |
|--------------------|----------------|
| A. 300.168.0.1 | B. 192.168.1.1 |
| C. 255.255.255.500 | D. 127.0.1.300 |

Correct Answer: B. 192.168.1.1

Detailed Option Breakdown: An IPv4 address is composed of 32 bits structured as 4 octets separated by dots (X . X . X . X). Each 8-bit octet must strictly be within the decimal range of **0 to 255** (00000000 to 11111111).

- **A. 300.168.0.1:** Invalid — the first octet 300 exceeds 255.
- **B. 192.168.1.1:** **[CORRECT] Valid — all four octets fall between 0 and 255 (standard Class C private IP).**
- **C. 255.255.255.500:** Invalid — the fourth octet 500 exceeds 255.
- **D. 127.0.1.300:** Invalid — the fourth octet 300 exceeds 255.

Q5 In which OSI layer does the HTTP protocol operate?

- | | |
|--------------|----------------|
| A. Transport | B. Network |
| C. Session | D. Application |

Correct Answer: D. Application

Detailed Option Breakdown:

- **A. Transport:** Layer 4 protocols include TCP and UDP.
- **B. Network:** Layer 3 protocols include IP, ICMP, ARP, and IGMP.
- **C. Session:** Layer 5 protocols include NetBIOS, RPC, and PPTP.
- **D. Application:** **[CORRECT] HTTP (HyperText Transfer Protocol) operates at Layer 7 (Application Layer) to facilitate data exchange between web browsers and servers.**

Layer	Name	PDU	Core Function & Operational Role	Key Protocols & Hardware
7	Application	Data	Interfaces directly with end-user applications and manages network services like web browsing and file transfers.	HTTP, HTTPS, FTP, SMTP, DNS, DHCP, Telnet, SSH
6	Presentation	Data	Translates data formats, handles data compression, and executes cryptographic encryption/decryption.	SSL/TLS, JPEG, PNG, ASCII, EBCDIC, MP4
5	Session	Data	Establishes, maintains, synchronizes, and terminates communication sessions between applications.	NetBIOS, RPC, PPTP, Sockets, Named Pipes
4	Transport	Segment / Datagram	Provides transparent, reliable end-to-end data transfer, error detection, retransmission, and flow control.	TCP, UDP, SCTP, Port Numbers (0–65535)
3	Network	Packet	Manages logical addressing, packet routing, and determination of optimal multi-hop internetwork paths.	IPv4, IPv6, ICMP, IPsec, Routers, Layer 3 Switches
2	Data Link	Frame	Provides node-to-node data transfer, framing, physical MAC addressing, and error detection (CRC).	Ethernet (802.3), Wi-Fi (802.11), Switches, Bridges
1	Physical	Bit	Transmits raw, unstructured bit streams over physical media (electrical pulses, optical signals, radio waves).	CAT6 Cables, Fiber Optics, Repeaters, Hubs

Q6 What does the term “latency” refer to in networking?

- A. The amount of data transmitted
- B. The time delay in data transmission
- C. The speed of the network
- D. The number of users on the network

Correct Answer: B. The time delay in data transmission

Detailed Option Breakdown:

- **A. The amount of data transmitted:** Refers to throughput or transferred data volume.
- **B. The time delay in data transmission:** **[CORRECT] Latency represents the total time delay incurred for a data packet to travel from its source to its destination (measured in milliseconds, ms).**
- **C. The speed of the network:** Bandwidth measures network channel capacity (e.g., Gbps).
- **D. The number of users:** Refers to concurrency or network traffic density.

Q7 What device is typically used to extend the range of a wireless network?

- A. Switch
- B. Router
- C. Repeater
- D. Bridge

Correct Answer: C. Repeater

Detailed Option Breakdown:

- **A. Switch:** Connects wired devices in a local network using MAC address tables.
- **B. Router:** Routes packets across different subnets.
- **C. Repeater (Range Extender):** **[CORRECT] A physical layer device that receives incoming wireless signals, amplifies/regenerates them, and retransmits them to eliminate dead zones.**
- **D. Bridge:** Connects two separate Layer 2 LAN segments.

Q8 In the OSI model, which layer is responsible for translating between different data formats?

- | | |
|-----------------|--------------|
| A. Presentation | B. Session |
| C. Network | D. Transport |

Correct Answer: A. Presentation

Detailed Option Breakdown:

- **A. Presentation (Layer 6):** **[CORRECT]** Acts as the network translator, standardizing character encodings (e.g., ASCII/EBCDIC), handling data compression, and executing encryption/decryption routines.
- **B. Session:** Establishes and manages dialogues between endpoints.
- **C. Network:** Handles routing and logical IP addressing.
- **D. Transport:** Ensures end-to-end packet delivery and flow control.

Q9 What type of network is restricted to a single building or campus?

- | | |
|--------|--------|
| A. WAN | B. LAN |
| C. PAN | D. MAN |

Correct Answer: B. LAN

Detailed Option Breakdown:

- **A. WAN (Wide Area Network):** Spans large geographic regions, countries, or the globe (e.g., the Internet).
- **B. LAN (Local Area Network):** **[CORRECT]** Confined to a geographically restricted zone such as a single home, office, school, or corporate campus.
- **C. PAN (Personal Area Network):** Centered around an individual within a range of a few meters (e.g., Bluetooth).
- **D. MAN (Metropolitan Area Network):** Covers an entire city or municipal area.

Q10 What technology is used to reduce collisions in Ethernet networks?

- | | |
|---------------|--------------|
| A. Token Ring | B. Hub |
| C. CSMA/CD | D. Broadcast |

Correct Answer: C. CSMA/CD

Detailed Option Breakdown:

- **A. Token Ring:** An older IEEE 802.5 token-passing ring architecture, not Ethernet.
- **B. Hub:** A shared multi-access device that amplifies collisions across all attached nodes.
- **C. CSMA/CD (Carrier Sense Multiple Access with Collision Detection):** **[CORRECT]** The media access protocol used in half-duplex Ethernet where nodes listen before transmitting, detect collisions, and invoke a random exponential backoff algorithm before retransmitting.
- **D. Broadcast:** Floods data to all devices, increasing collision likelihood.

Q11 To which class does the IP address 225.101.0.205 belong?

- A. Class A
- B. Class B
- C. Class C
- D. Class D

Correct Answer: D. Class D

Detailed Option Breakdown: IP classes are determined by analyzing the leading bits and range of the first octet (225):

- **A. Class A:** 1.0.0.0 to 126.255.255.255 (Leading bit pattern: 0).
- **B. Class B:** 128.0.0.0 to 191.255.255.255 (Leading bit pattern: 10).
- **C. Class C:** 192.0.0.0 to 223.255.255.255 (Leading bit pattern: 110).
- **D. Class D:** **[CORRECT] 224.0.0.0 to 239.255.255.255 (Leading bit pattern: 1110). 225 . 101 . 0 . 205 falls in this range and is strictly reserved for Multicast groups (e.g., video streams, routing protocol updates).**

Class	Public IP Range	Private IP Range	Subnet Mask	Designated Usage
Class A	1.0.0.0 – 126.255.255.255	10.0.0.0 – 10.255.255.255	255.0.0.0 (/8)	Very large enterprise networks
Class B	128.0.0.0 – 191.255.255.255	172.16.0.0 – 172.31.255.255	255.255.0.0 (/16)	Medium networks & universities
Class C	192.0.0.0 – 223.255.255.255	192.168.0.0 – 192.168.255.255	255.255.255.0 (/24)	Small businesses / SOHO
Class D	224.0.0.0 – 239.255.255.255	N/A (Reserved)	N/A	Multicasting (streaming media, OSPF)
Class E	240.0.0.0 – 255.255.255.255	N/A (Reserved)	N/A	Experimental & research purposes

2. NETWORK SECURITY

Q12 Which of the following is a form of two-factor authentication (2FA)?

- A. Password and CAPTCHA
- B. Password and fingerprint
- C. Username and password
- D. Password and PIN

Correct Answer: B. Password and fingerprint

Detailed Option Breakdown: True 2FA requires two distinct factors from: *Knowledge* (something you know), *Possession* (something you have), and *Inherence* (something you are).

- **A. Password and CAPTCHA:** CAPTCHA differentiates bots from humans; it is not an identity authentication factor.
- **B. Password and fingerprint:** **[CORRECT] Combines *Something you know* (Password) with *Something you are* (Biometric fingerprint).**
- **C. Username and password:** Single-factor authentication (Username is an identifier, password is knowledge).
- **D. Password and PIN:** Both belong to the same single category (*Something you know*).

Q13 What is the main purpose of a VPN (Virtual Private Network)?

- | | |
|----------------------------|--|
| A. Increase internet speed | B. Encrypt communication over public networks |
| C. Block malware | D. Prevent unauthorized users from accessing the LAN |

Correct Answer: B. Encrypt communication over public networks

Detailed Option Breakdown:

- **A. Increase internet speed:** Encryption and encapsulation overhead usually causes a minor reduction in throughput.
- **B. Encrypt communication over public networks:** **[CORRECT] Establishes a secure, encrypted tunnel over untrusted networks (the public Internet) to ensure confidentiality and integrity of transmitted data.**
- **C. Block malware:** Role of Antivirus and EDR software.
- **D. Prevent unauthorized users from accessing LAN:** Role of Firewalls and 802.1X Network Access Control (NAC).

Q14 Which of the following is considered a strong password policy?

- | | |
|--|--|
| A. Passwords containing only lowercase letters | B. Passwords of at least 8 characters, including symbols and numbers |
| C. Passwords that expire every 2 years | D. Passwords that use common dictionary words |

Correct Answer: B. Passwords of at least 8 characters, including symbols and numbers

Detailed Option Breakdown:

- **A. Only lowercase:** Low entropy; highly susceptible to quick brute-force attacks.
- **B. Minimum 8 characters with symbols & numbers:** **[CORRECT] High entropy derived from combining uppercase, lowercase, numerical digits, and special characters drastically increases resistance to cracking.**
- **C. Expire every 2 years:** Inadequate rotation cycle (industry standard is 60–90 days or event-based password resets).
- **D. Common dictionary words:** Highly vulnerable to dictionary attacks and rainbow table precomputed lookups.

Q15 Which protocol is commonly used to secure HTTP traffic?

- | | |
|--------|------------|
| A. SSH | B. SSL/TLS |
| C. FTP | D. IMAP |

Correct Answer: B. SSL/TLS

Detailed Option Breakdown:

- **A. SSH (Secure Shell):** Secures remote CLI terminal sessions (Port 22).
- **B. SSL/TLS (Transport Layer Security):** **[CORRECT] Encrypts HTTP traffic to form HTTPS (Port 443), providing data confidentiality, message integrity, and server authentication via digital certificates.**
- **C. FTP:** Insecure plaintext file protocol.
- **D. IMAP:** Email retrieval protocol (secured via IMAPS/TLS on Port 993).

Q16 What is a key feature of the “zero trust” security model?

- | | |
|-------------------------------|---|
| A. Trust but verify | B. Allow all traffic from internal networks |
| C. Always verify, never trust | D. Rely on firewalls alone |

Correct Answer: C. Always verify, never trust

Detailed Option Breakdown:

- **A. Trust but verify:** The legacy perimeter security concept, vulnerable once an attacker penetrates inside.
- **B. Allow all internal traffic:** Violates principle of least privilege and allows lateral threat propagation.
- **C. Always verify, never trust:** **[CORRECT] Core mantra of Zero Trust Architecture (ZTA). Eliminates implicit trust; strictly requires continuous authentication, dynamic authorization, and encryption for every user, device, and packet flow.**
- **D. Rely on firewalls alone:** Fails to protect against compromised credentials and insider threats.

Q17 What type of attack involves overwhelming a network or service with excessive traffic?

- | | |
|-------------|----------------------|
| A. Phishing | B. Man-in-the-Middle |
| C. DoS/DDoS | D. SQL Injection |

Correct Answer: C. DoS/DDoS

Detailed Option Breakdown:

- **A. Phishing:** Social engineering attack targeting credentials.
- **B. Man-in-the-Middle:** Eavesdropping and altering active transit communications.
- **C. DoS/DDoS (Distributed Denial-of-Service):** **[CORRECT] Floods a targeted network, server, or web application with high volumes of malicious traffic from single or distributed botnets to deplete resources and deny access to legitimate users.**
- **D. SQL Injection:** Injecting malicious SQL commands to access or manipulate database records.

Q18 A “honeypot” in network security is:

- | | |
|-----------------------------------|--|
| A. A type of encryption algorithm | B. A decoy system designed to lure attackers |
| C. A firewall rule | D. A malware scanner |

Correct Answer: B. A decoy system designed to lure attackers

Detailed Option Breakdown:

- **A. Encryption algorithm:** Honeypots do not perform cryptographic operations.
- **B. A decoy system designed to lure attackers:** **[CORRECT] An isolated, sacrificial trap containing simulated vulnerabilities deployed to deceive attackers, divert them from real assets, and analyze their exploitation techniques and tools.**
- **C. A firewall rule:** Incorrect definition.
- **D. A malware scanner:** Role of Antivirus/Endpoint Protection systems.

Q19 Which type of attack attempts to intercept and manipulate communication between two parties?

- | | |
|---------------|----------------------|
| A. Phishing | B. Man-in-the-Middle |
| C. Ransomware | D. Brute Force |

Correct Answer: B. Man-in-the-Middle

Detailed Option Breakdown:

- **A. Phishing:** Deceiving users into providing confidential information via spoofed communication.
- **B. Man-in-the-Middle (MitM):** **[CORRECT]** The attacker secretly positions themselves between two legitimate endpoints (e.g., via ARP poisoning or evil twin Wi-Fi) to eavesdrop, intercept, and alter data in transit without either party knowing.
- **C. Ransomware:** Extortion malware that encrypts files and demands ransom.
- **D. Brute Force:** Exhaustive password guessing.

Q20 What is the function of a DMZ in a network architecture?

- | | |
|---|---------------------------------------|
| A. To isolate internal resources from public access | B. To filter spam emails |
| C. To block malicious traffic | D. To encrypt internal communications |

Correct Answer: A. To isolate internal resources from public access

Detailed Option Breakdown:

- **A. To isolate internal resources from public access:** **[CORRECT]** A Demilitarized Zone (DMZ) is a perimeter subnet hosting external-facing servers (Web, DNS, Mail). If a DMZ host is breached, the internal private LAN remains shielded behind an inner firewall layer.
- **B. To filter spam emails:** Handled by Secure Email Gateways (SEG).
- **C. To block malicious traffic:** Handled by Firewalls and IPS.
- **D. To encrypt internal communications:** Handled by TLS/IPsec protocols.

Q21 Which of the following is a method of data encryption?

- | | |
|--------|---------|
| A. AES | B. SMTP |
| C. FTP | D. HTTP |

Correct Answer: A. AES

Detailed Option Breakdown:

- **A. AES (Advanced Encryption Standard):** **[CORRECT]** A symmetric block cipher utilized globally for robust data encryption.
- **B. SMTP:** Simple Mail Transfer Protocol (email delivery).
- **C. FTP:** File Transfer Protocol (unencrypted file transmission).
- **D. HTTP:** HyperText Transfer Protocol (unencrypted web transmission).

Q22 Which of the following is used for establishing a VPN connection?

- | | |
|------------|-------------|
| A. NGINX | B. ZeuS |
| C. HAProxy | D. Openswan |

Correct Answer: D. Openswan

Detailed Option Breakdown:

- **A. NGINX:** High-performance web server, reverse proxy, and Layer 7 load balancer.
- **B. ZeuS:** A well-known banking Trojan malware family used to steal credentials via keylogging.
- **C. HAProxy:** High-availability TCP/HTTP load balancer and proxy server.
- **D. Openswan:** **[CORRECT]** An open-source implementation of IPsec (Internet Protocol Security) for Linux systems, designed specifically for configuring site-to-site and remote-access VPN connections.

3. ENCRYPTION STANDARDS & ALGORITHMS

Q23 What does AES stand for in encryption?

- | | |
|-------------------------------------|---------------------------------|
| A. Asynchronous Encryption Standard | B. Advanced Encryption Standard |
| C. Algorithmic Encryption Scheme | D. Applied Encryption System |

Correct Answer: B. Advanced Encryption Standard

Detailed Option Breakdown:

- **B. Advanced Encryption Standard:** **[CORRECT]** A symmetric block cipher established by NIST in 2001 (based on the Rijndael algorithm) utilizing cryptographic key sizes of 128, 192, or 256 bits to secure classified and commercial data.
- **A, C, D:** Incorrect terminology.

Q24 Which encryption algorithm uses two different keys for encryption and decryption?

- | | |
|--------|-------------|
| A. DES | B. AES |
| C. RSA | D. Blowfish |

Correct Answer: C. RSA

Detailed Option Breakdown:

- **A. DES:** Symmetric cipher using a single 56-bit key.
- **B. AES:** Symmetric cipher using a single key (128/192/256 bits).
- **C. RSA (Rivest–Shamir–Adleman):** **[CORRECT]** An asymmetric encryption algorithm that generates a mathematically linked Public Key (for encryption) and Private Key (for decryption).
- **D. Blowfish:** Symmetric cipher using a variable-length single key.

Q25 What type of encryption uses the same key for both encryption and decryption?

- | | |
|--------------------------|--------------------------|
| A. Symmetric encryption | B. Asymmetric encryption |
| C. Public-key encryption | D. Hashing |

Correct Answer: A. Symmetric encryption

Detailed Option Breakdown:

- **A. Symmetric encryption:** [CORRECT] Uses a shared single secret key for both encrypting and decrypting data. It provides high-speed throughput for large volumes of data.
- **B. Asymmetric encryption:** Uses two mathematically related keys (public/private).
- **C. Public-key encryption:** Synonym for asymmetric encryption.
- **D. Hashing:** A one-way integrity check function, not encryption.

Q26 Which of the following encryption methods is considered outdated and vulnerable to attacks?

- | | |
|--------|--------|
| A. AES | B. RSA |
| C. DES | D. ECC |

Correct Answer: C. DES

Detailed Option Breakdown:

- **A. AES:** Highly secure modern symmetric cipher standard.
- **B. RSA:** Secure asymmetric algorithm when deployed with ≥ 2048 -bit keys.
- **C. DES (Data Encryption Standard):** [CORRECT] Outdated 56-bit symmetric cipher whose short key space can be broken in a matter of hours via brute-force. Officially superseded by AES.
- **D. ECC (Elliptic Curve Cryptography):** Modern asymmetric algorithm delivering equivalent security to RSA with much smaller key lengths.

Q27 What is the primary purpose of encryption in data security?

- | | |
|---|---|
| A. To compress data for faster transmission | B. To prevent unauthorized access to data |
| C. To ensure data integrity | D. To detect network intrusions |

Correct Answer: B. To prevent unauthorized access to data

Detailed Option Breakdown:

- **A. To compress data:** Role of compression codecs (e.g., Gzip, Deflate).
- **B. To prevent unauthorized access to data:** [CORRECT] Encryption provides confidentiality by transforming readable plaintext into unreadable ciphertext, ensuring only holders of the correct decryption key can read it.
- **C. To ensure data integrity:** Provided primarily by cryptographic Hashing (SHA-256) and MACs.
- **D. To detect intrusions:** Handled by IDS/IPS systems.

Q28 What is the difference between hashing and encryption?

- | | |
|--|---|
| A. Hashing is reversible; encryption is not | B. Encryption provides data integrity; hashing provides confidentiality |
| C. Hashing is irreversible; encryption is reversible | D. Encryption is faster than hashing |

Correct Answer: C. Hashing is irreversible; encryption is reversible

Detailed Option Breakdown:

- **A & B:** Factually inverted statements.
- **C. Hashing is irreversible; encryption is reversible: [CORRECT]** Hashing is a one-way mathematical transformation generating a fixed-length digest from variable inputs (cannot recover the original text). Encryption is a two-way function explicitly designed to allow decryption back to plaintext with the secret key.
- **D. Encryption is faster:** False; hashing functions are generally faster.

Q29 Which of the following algorithms is a hashing function?

- | | |
|------------|--------|
| A. AES | B. RSA |
| C. SHA-256 | D. ECC |

Correct Answer: C. SHA-256

Detailed Option Breakdown:

- **A. AES:** Symmetric block cipher.
- **B. RSA:** Asymmetric public-key encryption cipher.
- **C. SHA-256 (Secure Hash Algorithm 256-bit): [CORRECT]** A cryptographic hash algorithm belonging to the SHA-2 family that produces a fixed 256-bit fingerprint for data integrity and password hashing.
- **D. ECC:** Asymmetric public-key cipher based on elliptic curve mathematics.

Q30 In encryption, what is a “key”?

- | | |
|--|--|
| A. A device used to encrypt files | B. A random string used to encrypt or decrypt data |
| C. A mathematical formula that compresses data | D. A digital certificate used to verify identity |

Correct Answer: B. A random string used to encrypt or decrypt data

Detailed Option Breakdown:

- **A. A device:** Keys can be stored on hardware (HSM/smartcard), but the key itself is mathematical data.
- **B. A random string used to encrypt or decrypt data: [CORRECT]** A binary parameter/string supplied into a cryptographic algorithm to govern encryption and decryption.
- **C. A formula:** The formula is the *cipher/algorithm*; the key is the variable input.
- **D. A digital certificate:** Certificates wrap and bind a public key to an identity.

Q31 What does the term “public key” in encryption refer to?

- A. A key that encrypts data but cannot decrypt it
- B. A key that is shared with everyone for encryption purposes
- C. A key used to generate random numbers
- D. A key that must remain private to the user

Correct Answer: B. A key that is shared with everyone for encryption purposes

Detailed Option Breakdown:

- **A. Cannot decrypt:** In digital signature schemes, the public key is used to decrypt/verify signatures created by the private key.
- **B. A key that is shared with everyone for encryption purposes:** **[CORRECT]** In asymmetric setups, the public key is openly distributed so senders can encrypt data that only the matching private key can decrypt.
- **C. Generates random numbers:** Incorrect.
- **D. Must remain private:** This describes the confidential **Private Key**.

Q32 Which encryption algorithm is based on the difficulty of factoring large numbers?

- A. RSA
- B. AES
- C. DES
- D. ECC

Correct Answer: A. RSA

Detailed Option Breakdown:

- **A. RSA:** **[CORRECT]** Security relies directly on the *Prime Factorization Problem* (multiplying two large prime numbers is easy, but factoring their large composite product back into primes is computationally infeasible for modern computers).
- **B & C. AES / DES:** Based on Substitution-Permutation Networks and Feistel structures.
- **D. ECC:** Relies on the Discrete Logarithm Problem over elliptic curves.

4. NETWORK SECURITY DEVICES

Q33 Which device is primarily used to prevent unauthorized access to or from a private network?

- A. Switch
- B. Firewall
- C. Router
- D. Hub

Correct Answer: B. Firewall

Detailed Option Breakdown:

- **A. Switch:** Forwards local frames using MAC addresses.
- **B. Firewall:** **[CORRECT]** Inspects inbound and outbound network traffic against configured security rules to prevent unauthorized perimeter access.
- **C. Router:** Routes packets between different IP subnets.
- **D. Hub:** Broadcasts raw electrical signals to all connected physical ports.

Q34 What is the function of an Intrusion Detection System (IDS)?

- | | |
|-------------------------------|--|
| A. Block unauthorized traffic | B. Monitor network traffic for suspicious activity |
| C. Encrypt network data | D. Route network traffic |

Correct Answer: B. Monitor network traffic for suspicious activity

Detailed Option Breakdown:

- **A. Block unauthorized traffic:** Role of an inline **IPS (Intrusion Prevention System)**.
- **B. Monitor network traffic for suspicious activity:** **[CORRECT] A passive (out-of-band) security system that inspects mirrored packets against threat signatures and behavioral baselines, generating real-time alerts.**
- **C. Encrypt network data:** Function of VPNs/IPsec gateways.
- **D. Route network traffic:** Function of Layer 3 routers.

Q35 Which device can automatically block suspicious traffic after detecting it?

- | | |
|-----------|------------------|
| A. IDS | B. IPS |
| C. Router | D. Load balancer |

Correct Answer: B. IPS

Detailed Option Breakdown:

- **A. IDS:** Passive alerting only; does not sit inline to drop traffic.
- **B. IPS (Intrusion Prevention System):** **[CORRECT] Deployed inline with the network flow; automatically blocks malicious packets, resets connections, and alters firewall rules in real time.**
- **C. Router:** Routes traffic without deep attack inspection.
- **D. Load balancer:** Balances server workload distributions.

Q36 Which network security device acts as a proxy and inspects HTTPS traffic?

- | | |
|-----------------------------------|-----------|
| A. Web Application Firewall (WAF) | B. Router |
| C. IDS | D. VPN |

Correct Answer: A. Web Application Firewall (WAF)

Detailed Option Breakdown:

- **A. Web Application Firewall (WAF):** **[CORRECT] Operates at Layer 7 as a reverse proxy, decrypting and inspecting HTTPS payloads to detect attacks like SQLi, XSS, CSRF, and session hijacking.**
- **B. Router:** Operates at Layer 3; cannot parse application web payloads.
- **C. IDS:** Lower-layer packet monitoring.
- **D. VPN:** Encrypts tunnels; does not inspect HTTP payloads.

Q37 What is the main function of a load balancer in network security?

- | | |
|---|------------------------------------|
| A. Distribute traffic across multiple servers | B. Encrypt traffic between servers |
| C. Filter incoming traffic | D. Detect and prevent intrusions |

Correct Answer: A. Distribute traffic across multiple servers

Detailed Option Breakdown:

- **A. Distribute traffic across multiple servers:** [CORRECT] Balances incoming user requests across a pool of backend servers to prevent server overload, maximize availability, and eliminate single points of failure.
- **B. Encrypt traffic:** Handled by TLS termination.
- **C. Filter traffic:** Role of Firewalls.
- **D. Detect/prevent intrusions:** Role of IDS/IPS.

Q38 Which device is often deployed to analyze and filter out email-based attacks?

- | | |
|-------------|----------------------|
| A. Firewall | B. Anti-spam gateway |
| C. Router | D. Switch |

Correct Answer: B. Anti-spam gateway

Detailed Option Breakdown:

- **A. Firewall:** Filters IP ports (e.g., port 25) without in-depth email body and attachment analysis.
- **B. Anti-spam gateway (Secure Email Gateway - SEG):** [CORRECT] Inspects inbound/outbound SMTP traffic for phishing links, malicious attachments, spam, and spoofed sender addresses (using SPF, DKIM, DMARC).
- **C. Router:** Routes packets.
- **D. Switch:** Forwards local frames.

Q39 What does a DMZ (Demilitarized Zone) in network security typically contain?

- | | |
|--------------------------|------------------------------------|
| A. Public-facing servers | B. Confidential internal databases |
| C. End-user devices | D. Firewalls and routers |

Correct Answer: A. Public-facing servers

Detailed Option Breakdown:

- **A. Public-facing servers:** [CORRECT] Contains services needing external Internet access (Web, External DNS, Mail) while strictly isolating internal private resources.
- **B. Confidential databases:** Kept in deep internal secure database zones.
- **C. End-user devices:** Located inside the internal corporate LAN.
- **D. Firewalls and routers:** Form the perimeter boundaries that establish the DMZ.

Q40 A UTM (Unified Threat Management) device is designed to:

- | | |
|---|---|
| A. Block all incoming traffic | B. Provide a comprehensive set of security features in one device |
| C. Control routing of traffic between subnets | D. Monitor internal data breaches |

Correct Answer: B. Provide a comprehensive set of security features in one device

Detailed Option Breakdown:

- **A. Block all traffic:** Insecure baseline misconfiguration.
- **B. Comprehensive security features in one device:** **[CORRECT] Integrates multiple security functions (Stateful Firewall, Antivirus, IDS/IPS, URL/Content Filtering, Anti-Spam, and VPN) into a single unified hardware appliance.**
- **C. Control routing:** Handled by routers.
- **D. Monitor internal breaches:** Role of SIEM and DLP systems.

Q41 Which of the following is an example of a stateful firewall?

- | | |
|--|---|
| A. A firewall that only examines IP addresses | B. A firewall that tracks the state of active connections |
| C. A firewall that allows all incoming traffic | D. A firewall that blocks all traffic by default |

Correct Answer: B. A firewall that tracks the state of active connections

Detailed Option Breakdown:

- **A. Only examines IP addresses:** Describes a stateless packet filter or simple ACL.
- **B. Tracks active connection state:** **[CORRECT] Maintains a state table tracking ongoing TCP/UDP sessions, automatically allowing inbound response traffic matching an established outbound request.**
- **C. Allows all traffic:** Insecure open policy.
- **D. Blocks all by default:** Describes an implicit deny policy, not connection state tracking.

Q42 Which device is typically used to connect multiple VLANs (Virtual Local Area Networks)?

- | | |
|-------------|------------------|
| A. Router | B. Switch |
| C. Firewall | D. Load Balancer |

Correct Answer: A. Router

Detailed Option Breakdown:

- **A. Router:** **[CORRECT] VLANs isolate networks at Layer 2 (separate broadcast domains). Routing traffic between different VLANs requires a Layer 3 routing device (or a Layer 3 Switch using Inter-VLAN routing / "Router-on-a-Stick").**
- **B. Switch:** Standard Layer 2 switches isolate VLANs and cannot route between them without Layer 3 routing logic.
- **C. Firewall:** Can route between zones, but routers/Layer 3 switches are the primary network architectural standard.
- **D. Load Balancer:** Distributes application traffic across servers.

5. FIREWALLS

Q43 What is the main purpose of a firewall in network security?

- A. Encrypt network traffic
- B. Block unauthorized access to or from a private network
- C. Distribute network traffic
- D. Monitor traffic for threats

Correct Answer: B. Block unauthorized access to or from a private network

Detailed Option Breakdown:

- **A. Encrypt traffic:** Handled by IPsec/TLS protocols.
- **B. Block unauthorized access:** **[CORRECT]** Enforces perimeter boundary security policies by inspecting incoming and outgoing packets against predefined security rules.
- **C. Distribute traffic:** Role of load balancers.
- **D. Monitor traffic:** Primary function of IDS and network monitoring tools.

Q44 Which type of firewall operates at the application layer of the OSI model?

- A. Packet-filtering firewall
- B. Stateful firewall
- C. Application-layer firewall
- D. Circuit-level gateway

Correct Answer: C. Application-layer firewall

Detailed Option Breakdown:

- **A. Packet-filtering:** Operates at Layers 3 and 4 (IP/Port).
- **B. Stateful firewall:** Operates across Layers 3 and 4, tracking connection state.
- **C. Application-layer firewall (Proxy / Layer 7):** **[CORRECT]** Understands specific application protocols (HTTP, FTP, DNS) and inspects the actual payload for malicious commands, unauthorized requests, and protocol violations.
- **D. Circuit-level gateway:** Operates at Layer 5, validating TCP handshakes without inspecting application payloads.

Q45 What is the difference between a stateful firewall and a stateless firewall?

- A. Stateless firewalls track the state of network connections
- B. Stateful firewalls remember the state of active connections
- C. Stateful firewalls do not filter traffic
- D. Stateless firewalls encrypt network traffic

Correct Answer: B. Stateful firewalls remember the state of active connections

Detailed Option Breakdown:

- **A. Stateless tracks state:** False. Stateless firewalls evaluate every packet independently without session context.
- **B. Stateful firewalls remember active connections:** **[CORRECT]** Maintains a dynamic state table. If a host sends an outbound request, returning packets belonging to that established session are automatically validated and allowed back in.
- **C. Stateful firewalls do not filter:** False. Filtering is their core function.
- **D. Stateless firewalls encrypt:** False. Neither type encrypts traffic by default.

Q46 Which type of firewall examines the data content (payload) of packets?

- | | |
|------------------------------|--|
| A. Packet-filtering firewall | B. Deep packet inspection (DPI) firewall |
| C. Circuit-level firewall | D. Stateless firewall |

Correct Answer: B. Deep packet inspection (DPI) firewall

Detailed Option Breakdown:

- **A. Packet-filtering:** Inspects only basic packet headers (IP and Port).
- **B. Deep packet inspection (DPI) firewall (Next-Gen Firewall - NGFW):** **[CORRECT] Inspects past Layer 3/4 headers down into the actual data payload to uncover hidden malware, unauthorized protocols, and sensitive data leaks.**
- **C. Circuit-level:** Validates session connection handshakes only.
- **D. Stateless firewall:** Evaluates isolated packet headers against static rules.

Q47 A firewall rule configured to block all inbound traffic except HTTP and HTTPS would be an example of:

- | | |
|-----------------------|--------------------------------|
| A. Implicit deny | B. Explicit allow |
| C. Stateful filtering | D. Application-layer filtering |

Correct Answer: B. Explicit allow

Detailed Option Breakdown:

- **A. Implicit deny:** The underlying baseline rule that drops unmatched traffic, but the question describes creating an open rule for Ports 80 and 443.
- **B. Explicit allow:** **[CORRECT] A defined rule that specifically permits designated traffic (e.g., ALLOW TCP Ports 80, 443), while leaving remaining traffic to be caught by drop policies.**
- **C. Stateful filtering:** The mechanism of tracking connections, not the rule configuration syntax itself.
- **D. Application-layer filtering:** Content inspection, whereas port allowances are configured via port rules.

Q48 In what scenario would a firewall be configured in a “deny-all” mode?

- | | |
|--|---|
| A. When allowing unrestricted network access | B. When traffic is only allowed if explicitly permitted |
| C. When all traffic is filtered based on content | D. When ensuring performance optimization |

Correct Answer: B. When traffic is only allowed if explicitly permitted

Detailed Option Breakdown:

- **A. Unrestricted access:** Opposite of secure firewall administration.
- **B. Traffic allowed only if explicitly permitted:** **[CORRECT] Represents the Default Deny / Whitelisting security posture where all incoming and outgoing network traffic is blocked by default; administrators must create explicit allow rules for required services.**
- **C. Content-filtered only:** Incomplete policy definition.
- **D. Performance optimization:** Running open networks reduces compute overhead but leaves systems unprotected.

Q49 Which type of firewall operates by analyzing the headers of individual packets?

- | | |
|-------------------------------|------------------------------|
| A. Application-layer firewall | B. Packet-filtering firewall |
| C. Stateful firewall | D. Proxy firewall |

Correct Answer: B. Packet-filtering firewall

Detailed Option Breakdown:

- **A. Application-layer:** Evaluates full payloads and application commands.
- **B. Packet-filtering firewall:** **[CORRECT] The simplest firewall architecture; inspects individual packet headers (Source IP, Destination IP, Protocol, Source Port, Destination Port) against static Access Control Lists.**
- **C. Stateful firewall:** Evaluates active session state across flows, not just individual isolated packet headers.
- **D. Proxy firewall:** Terminates connections and establishes a new session on behalf of the client.

Q50 What is a “proxy firewall”?

- | | |
|---|--|
| A. A firewall that inspects and filters traffic at the network layer | B. A firewall that acts as an intermediary between clients and servers |
| C. A firewall that prevents internal users from accessing external websites | D. A firewall that blocks all incoming connections |

Correct Answer: B. A firewall that acts as an intermediary between clients and servers

Detailed Option Breakdown:

- **A. Operates at network layer:** Describes packet filters.
- **B. Acts as an intermediary between clients and servers:** **[CORRECT] Direct client-to-server network connections are prevented. The proxy terminates the client connection, inspects the requested payload, and opens a separate connection to the destination server on the client's behalf.**
- **C. Prevents web access:** This is URL blacklisting, which is just one feature a proxy can perform.
- **D. Blocks all incoming:** Describes a default deny policy.

Q51 Which firewall architecture is often used to protect internal networks from external threats while allowing public access to certain resources like web servers?

- | | |
|--------|----------------------|
| A. DMZ | B. Stateful firewall |
| C. VPN | D. NAT |

Correct Answer: A. DMZ

Detailed Option Breakdown:

- **A. DMZ (Demilitarized Zone):** **[CORRECT] A dual-firewall or multi-homed firewall layout that separates public-facing servers from the internal private LAN, ensuring external web users cannot directly route into internal networks.**
- **B. Stateful firewall:** A filtering engine, not a structural network zoning topology.
- **C. VPN:** An encrypted tunnel for remote access.
- **D. NAT (Network Address Translation):** Translates private IP addresses to public IPs, but does not provide multi-tiered subnet isolation by itself.

Q52 In firewall rule configuration, what does “implicit deny” mean?

- A. All traffic is allowed unless explicitly blocked
- B. All traffic is blocked unless explicitly allowed
- C. Traffic from external sources is always blocked
- D. Traffic is encrypted by default

Correct Answer: B. All traffic is blocked unless explicitly allowed

Detailed Option Breakdown:

- **A. Allowed unless blocked:** Describes an insecure "Default Allow" / Blacklist design.
- **B. Blocked unless explicitly allowed: [CORRECT]** The foundational security principle placed at the bottom of access lists: if incoming or outgoing traffic does not match any specific allow rule above it, it is automatically dropped.
- **C. External traffic always blocked:** Inaccurate, as legitimate external traffic can be permitted via explicit rules.
- **D. Encrypted by default:** Firewalls filter traffic rather than encrypting all packets.

6. ATTACK TYPES

Q53 What is the primary goal of a ransomware attack?

- A. To disable a system with excessive traffic
- B. To steal sensitive data
- C. To encrypt a victim's data and demand payment for decryption
- D. To exploit a software vulnerability for remote access

Correct Answer: C. To encrypt a victim's data and demand payment for decryption

Detailed Option Breakdown:

- **A. Disable system with traffic:** Describes a DoS/DDoS attack.
- **B. Steal sensitive data:** Describes Data Exfiltration / Espionage (though double-extortion ransomware does this, encryption for ransom is the defining characteristic).
- **C. Encrypt data and demand payment: [CORRECT]** Ransomware uses strong asymmetric/symmetric encryption to lock user files, databases, or operating systems, demanding cryptocurrency payment for the private decryption key.
- **D. Exploit for remote access:** Describes Trojans, RATs, or Backdoors.

Q54 In which attack does an attacker try all possible password combinations to gain unauthorized access?

- A. Phishing
- B. SQL Injection
- C. Brute Force
- D. Man-in-the-Middle

Correct Answer: C. Brute Force

Detailed Option Breakdown:

- **A. Phishing:** Social engineering to trick users into revealing credentials willingly.
- **B. SQL Injection:** Database payload injection.
- **C. Brute Force: [CORRECT]** An exhaustive mathematical attack where automated software iterates through every possible character combination (a, b, c... 1, 2, 3...) until the correct password matches.
- **D. Man-in-the-Middle:** Intercepting live communications.

Q55 What type of attack targets vulnerabilities in web applications by injecting malicious code into a database query?

- | | |
|------------------|-------------------------------|
| A. Phishing | B. Cross-Site Scripting (XSS) |
| C. SQL Injection | D. Denial-of-Service |

Correct Answer: C. SQL Injection

Detailed Option Breakdown:

- **A. Phishing:** Social engineering via email or fraudulent websites.
- **B. Cross-Site Scripting (XSS):** Injects malicious client-side JavaScript into web pages viewed by other users.
- **C. SQL Injection (SQLi):** **[CORRECT]** Exploits unsanitized web form input fields to inject arbitrary SQL statements (e.g., ' OR '1'='1'), allowing attackers to bypass authentication, dump tables, or delete data.
- **D. Denial-of-Service:** Flooding infrastructure to make it unavailable.

Q56 Which type of attack redirects traffic from a legitimate site to a malicious one, usually by compromising DNS servers?

- | | |
|-----------------|----------------------|
| A. DNS Spoofing | B. Man-in-the-Middle |
| C. Phishing | D. SQL Injection |

Correct Answer: A. DNS Spoofing

Detailed Option Breakdown:

- **A. DNS Spoofing (DNS Cache Poisoning):** **[CORRECT]** Attackers corrupt a DNS resolver's cache with fraudulent IP mappings, silently redirecting users requesting legitimate domain names (e.g., bank.com) to an attacker-controlled server.
- **B. Man-in-the-Middle:** Eavesdropping/altering data in transit between two endpoints.
- **C. Phishing:** Deceiving users into clicking fake links via fraudulent messages.
- **D. SQL Injection:** Database exploitation technique.

Q57 What is the main goal of a social engineering attack?

- | | |
|--|--|
| A. To exploit software vulnerabilities | B. To trick individuals into revealing sensitive information |
| C. To flood a network with traffic | D. To execute malicious code on a server |

Correct Answer: B. To trick individuals into revealing sensitive information

Detailed Option Breakdown:

- **A. Exploit software vulnerabilities:** Technical exploitation (buffer overflows, zero-days).
- **B. Trick individuals into revealing sensitive info:** **[CORRECT]** Targets human psychology (authority, urgency, fear) to manipulate people into breaking security procedures, sharing passwords, or granting unauthorized physical access.
- **C. Flood network:** Network DoS attack.
- **D. Execute malicious code:** Remote Code Execution (RCE).

Q58 What is a “buffer overflow” attack?

- | | |
|---|--|
| A. An attack that overwhelms a network with traffic | B. An attack that exploits a vulnerability by sending more data than a buffer can handle |
| C. An attack that targets weak passwords | D. An attack that encrypts the victim's files |

Correct Answer: B. An attack that exploits a vulnerability by sending more data than a buffer can handle

Detailed Option Breakdown:

- **A. Overwhelms network:** Network Denial of Service.
- **B. Sends more data than a buffer can handle:** **[CORRECT]** Occurs when an application writes more data to a memory buffer than allocated. The excess data overwrites adjacent memory spaces (like the Instruction Pointer, EIP/RIP), allowing attackers to alter execution flow and inject arbitrary shellcode.
- **C. Targets weak passwords:** Brute-force / Dictionary attacks.
- **D. Encrypts files:** Ransomware.

Q59 What type of malware restricts access to data or systems until a ransom is paid?

- | | |
|---------------|------------|
| A. Trojan | B. Worm |
| C. Ransomware | D. Spyware |

Correct Answer: C. Ransomware

Detailed Option Breakdown:

- **A. Trojan:** Malware disguised as legitimate software that secretly carries a malicious payload.
- **B. Worm:** Standalone self-replicating malware that spreads across networks without user intervention.
- **C. Ransomware:** **[CORRECT]** Software specifically designed to deny access to a computer system or data by encrypting files until a ransom payment is submitted.
- **D. Spyware:** Secretly monitors and gathers user activity, credentials, and browsing patterns.

Q60 What kind of attack involves repeatedly sending ping requests to a network to overwhelm the target system?

- | | |
|----------------------|------------------|
| A. DDoS | B. SQL Injection |
| C. Man-in-the-Middle | D. Trojan |

Correct Answer: A. DDoS

Detailed Option Breakdown:

- **A. DDoS (Distributed Denial-of-Service / Ping Flood / ICMP Flood):** **[CORRECT]** Overwhelms the target's network bandwidth and CPU processing queue by continuously flooding it with ICMP Echo Request (ping) packets from multiple compromised systems.
- **B. SQL Injection:** Database exploitation method.
- **C. Man-in-the-Middle:** Communication eavesdropping.
- **D. Trojan:** Concealed malicious program.

Q61 In a phishing attack, what is typically impersonated to trick users?

- A. A hardware device
- B. A legitimate website or service
- C. A software application
- D. A local network

Correct Answer: B. A legitimate website or service

Detailed Option Breakdown:

- **A. Hardware device:** Inaccurate for traditional phishing delivery.
- **B. A legitimate website or service:** **[CORRECT]** Attackers craft spoofed emails, domain typos (typosquatting), and cloned landing pages mimicking trusted organizations (e.g., Microsoft, Google, PayPal) to trick users into entering credentials.
- **C & D:** While software updates or network alerts can be simulated, the attack relies on impersonating trusted organizations, brands, or service platforms.

Q62 What type of attack exploits a vulnerability that is unknown to the software vendor and has no fix available?

- A. Zero-Day Attack
- B. Man-in-the-Middle
- C. DDoS
- D. Trojan

Correct Answer: A. Zero-Day Attack

Detailed Option Breakdown:

- **A. Zero-Day Attack:** **[CORRECT]** Exploits a software vulnerability that is unknown to the vendor and security community. Because the vendor has had "zero days" to develop and release a security patch, defenses rely on behavioral heuristics and anomaly detection.
- **B. Man-in-the-Middle:** Network interception method.
- **C. DDoS:** Resource exhaustion method.
- **D. Trojan:** Malware delivery mechanism.

7. COMPREHENSIVE SUMMARY & COMPARISON TABLES

Comparison Parameter	Symmetric Encryption	Asymmetric Encryption
Number of Keys	1 (Single shared secret key)	2 (Mathematically linked Public & Private key pair)
Execution Speed	Extremely fast with low computational load	Slower, highly computation/math intensive
Key Distribution	Challenging (requires secure pre-shared exchange)	Straightforward (Public key is shared openly)
Core Use Cases	Bulk data storage, database and disk encryption	Key exchange, digital signatures, TLS handshakes
Standard Algorithms	AES (128/192/256), DES, 3DES, Blowfish, ChaCha20	RSA (2048/4096), ECC (ECDSA, Ed25519), Diffie-Hellman

Attack Type	Target / Vector	Operational Impact	Standard Industry Defense
SQL Injection (SQLi)	Database query execution	Data breaches, privilege escalation, data destruction	Prepared statements (Parameterized queries), ORMs, WAF
Cross-Site Scripting (XSS)	Client-side browser execution	Session hijacking, cookie theft, DOM manipulation	Context-aware output encoding, Content Security Policy (CSP)
DDoS Flooding	Network bandwidth, CPU/memory	Service outage, total unavailability	Anycast routing, Cloud Scrubbing centers, Rate limiting
Man-in-the-Middle (MitM)	Active network transit lines	Data eavesdropping, session modification	End-to-end TLS encryption, Dynamic ARP Inspection, HSTS
Buffer Overflow	System memory (Stack/Heap)	Arbitrary code execution, system takeover	Bounds checking, ASLR, DEP/NX Bit, memory-safe languages
Zero-Day Exploit	Unpatched code vulnerabilities	Unrestricted system penetration	Behavioral Endpoint Detection (EDR), Sandboxing, Zero Trust